

NETWORK PROTOCOLS AND SECURITY – LAB MANUAL – 2025-26

Switch-4	Switch-5
<pre>Switch>en Switch#config t Switch(config)#vlan 10 Switch(config-vlan)#name Sales Switch(config-vlan)#vlan 20 Switch(config-vlan)#name Admin Switch(config-vlan)#exit Switch(config)#int f0/1 Switch(config-if)#switchport mode access Switch(config-if)#switchport access vlan 10 Switch(config-if)#exit Switch(config)#int f0/2 Switch(config-if)#switchport mode access Switch(config-if)#switchport access vlan 20 Switch(config)#int g0/1 Switch(config-if)#switchport mode trunk Switch(config)#int g0/2 Switch(config-if)#switchport mode trunk</pre>	<pre>Switch>en Switch#config t Switch(config)#vlan 10 Switch(config-vlan)#name Sales Switch(config-vlan)#vlan 20 Switch(config-vlan)#name Admin Switch(config-vlan)#exit Switch(config)#int f0/1 Switch(config-if)#switchport mode access Switch(config-if)#switchport access vlan 10 Switch(config-if)#exit Switch(config)#int f0/2 Switch(config-if)#switchport mode access Switch(config-if)#switchport access vlan 20 Switch(config)#int g0/1 Switch(config-if)#switchport mode trunk</pre>

Configuration of Multilayer Switch

```
Switch>en
Switch#config t
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Admin
Switch(config-vlan)#exit
Switch(config)#int vlan 10
Switch(config-if)#int vlan 10
Switch(config-if)#ip address 10.0.0.1 255.0.0.0
Switch(config-if)#exit
Switch(config)#int vlan 20
Switch(config-if)#ip address 11.0.0.1 255.0.0.0
Switch(config-if)#exit
Switch(config)#int g1/0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#exit
Switch(config)#ip routing
```

Configuration of PCs

```
PC10: IP Address- 10.0.0.2
      Subnet Mask- 255.0.0.0
      Default Gateway- 10.0.0.1
PC11: IP Address- 11.0.0.2
      Subnet Mask- 255.0.0.0
      Default Gateway- 11.0.0.1
PC12: IP Address- 10.0.0.3
      Subnet Mask- 255.0.0.0
      Default Gateway- 10.0.0.1
PC13: IP Address- 11.0.0.3
      Subnet Mask- 255.0.0.0
      Default Gateway- 11.0.0.1
```

Conclusion

This configuration sets up 802.1Q encapsulation on trunk ports, allowing VLAN traffic to be carried across a single link between switches. By following these steps, you can manage multiple VLANs efficiently within your network using Cisco Packet Tracer.

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